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Learning Occlusion-Robust Vision Transformers for Real-Time UAV Tracking

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Abstract

Single-stream architectures using Vision Transformer (ViT) backbones show great potential for real-time UAV tracking recently. However, frequent occlusions from obstacles like buildings and trees expose a major drawback: these models often lack strategies to handle occlusions effectively. New methods are needed to enhance the occlusion resilience of single-stream ViT models in aerial tracking. In this work, we propose to learn Occlusion-Robust Representations (ORR) based on ViTs for UAV tracking by enforcing an invariance of the feature representation of a target with respect to random masking operations modeled by a spatial Cox process. Hopefully, this random masking approximately simulates target occlusions, thereby enabling us to learn ViTs that are robust to target occlusion for UAV tracking. This framework is termed ORTrack. Additionally, to facilitate real-time applications, we propose an Adaptive Feature-Based Knowledge Distillation (AFKD) method to create a more compact tracker, which adaptively mimics the behavior of the teacher model ORTrack according to the task's difficulty. This student model, dubbed ORTrack-D, retains much of ORTrack's performance while offering higher efficiency. Extensive experiments on multiple benchmarks validate the effectiveness of our method, demonstrating its state-of-the-art performance. Codes is available at https://github.com/wuyou3474/ORTrack.

1. Introduction

Unmanned aerial vehicles (UAVs) are leveraged in a plethora of applications, with increasing emphasis on UAV tracking [4, 43, 46, 49, 52, 79, 84]. This form of tracking poses an exclusive set of challenges such as tricky viewing angles, motion blur, severe occlusions, and the need for efficiency due to UAVs' restricted battery life and computational resources [5, 41, 80, 83]. Consequently, designing an effective UAV tracker requires a delicate balance between precision and efficiency. It needs to ensure accuracy while being conscious of the UAV's energy and computational constraints.

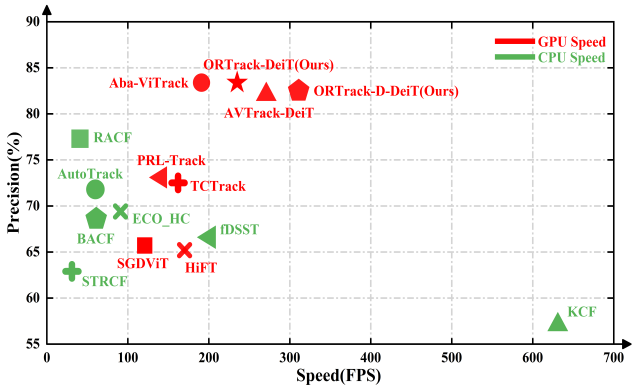


Figure 1. Compared to SOTA UAV trackers on UAVDT, our ORTrack-DeiT sets a new record with 83.4% precision and a speed of 236 FPS. Our ORTrack-D-DeiT strikes a better trade-off with 82.5% precision and a speed of about 313 FPS.

In recent years, there has been a notable shift from discriminative correlation filters (DCF)-based methods, because of their unsatisfactory robustness, towards DL-based approaches, particularly with the adoption of single-stream architectures that integrate feature extraction and fusion via pre-trained Vision Transformer (ViT) backbone networks. This single-stream paradigm has proven highly effective in generic visual tracking, as evidenced by the success of recent methods such as OSTrack [91], SimTrack [8], Mixformer [13], and DropMAE [82]. Building on these advancements, Aba-VTrack [44] introduces a lightweight DL-based tracker within this framework, employing an adap-

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tive and background-aware token computation method to enhance inference speed, which demonstrates remarkable precision and speed for real-time UAV tracking. However, the use of a variable number of tokens in Aba-VTrack incurs significant time costs, primarily due to the unstructured access operations required during inference. Adding to this, it also grappled with establishing robustness when facing target occlusion, a challenge common in UAV tracking often triggered by obstructive elements like buildings, mountains, trees, and so forth. The problem is exacerbated by the fact that UAVs may not always be capable of circumventing these impediments due to potential large-scale movements involved.

To address these issues, we introduce a novel framework designed to enhance the occlusion robustness of ViTs for UAV tracking. Our approach, termed ORTrack, aims to learn ViT-based trackers that maintain robust feature representations even in the presence of target occlusion. This is achieved by enforcing an invariance in the feature representation of the target with respect to random masking operations modeled by a spatial Cox process. The random masking serves as a simulation of target occlusion, which is expected to mimic real occlusion challenges in UAV tracking and aid in learning Occlusion-Robust Representations (ORR). Notably, our method for learning occlusion-robust representation simply uses a Mean Squared Error (MSE) loss during training, adding no extra computational load during inference. Additionally, to enhance efficiency for real-time applications, we introduce an Adaptive Feature-Based Knowledge Distillation (AFKD) method. This method creates a more compact tracker, named ORTrack-D, which adaptively mimics the behavior of the teacher model ORTrack based on the complexity of the tracking task during training. The reasoning is that the teacher model, in its pursuit of powerful representations, may compromise its generalizability. Hence, in situations where generalizability is vital, the student model may perform better, and closely mimicking the teacher's behavior becomes less important. We use the deviation of GIoU loss [67] from its average value to quantify the difficulty of the tracking task, which makes sense as loss value is a commonly used criteria to define hard samples [70, 74, 77]. ORTrack-D maintains much of ORTrack's performance with higher efficiency, making it better suited for deployment in resource-constrained environments typical of UAV applications. Extensive experiments on four benchmarks show that our method achieves state-of-the-art performance.

In summary, our contributions are as follows: (i) We propose to learn Occlusion-Robust Representations (ORR) by imposing an invariance in the feature representation of the target with respect to random masking operations modeled by a spatial Cox process, which can be easily integrated into other tracking frameworks without requiring additional architectures or increasing inference time; (ii) We propose an Adaptive Feature-Based Knowledge Distillation (AFKD) method to further enhance efficiency, in which the student model adaptively mimics the behavior of the teacher model according to the task's difficulty, resulting in a significant increase in tracking speed while only minimally reducing accuracy; (iii) We introduce ORTrack, a family of efficient trackers based on these components, which integrates seamlessly with other ViT-based trackers. ORTrack demonstrates superior performance while maintaining extremely fast tracking speeds. Extensive evaluations show that ORTrack achieves state-of-the-art real-time performance.

2. Related work

2.1. Visual Tracking.

In visual tracking, the primary approaches consist of DCF-based and DL-based trackers. DCF-based trackers are favored for UAV tracking due to their remarkable efficiency, but they face difficulties in maintaining robustness under complex conditions [31, 41, 46]. Recently developed lightweight DL-based trackers have improved tracking precision and robustness for UAV tracking [4, 5]; however, their efficiency lags behind that of most DCF-based trackers. Model compression techniques like those in [80, 83] have been used to further boost efficiency, yet these trackers still face issues with tracking precision. Vision Transformers (ViTs) are gaining traction for streamlining and unifying frameworks in visual tracking, as seen in studies like [13, 85, 86, 89, 91]. While these frameworks are compact and efficient, few are based on lightweight ViTs, making them impractical for real-time UAV tracking. To address this, Aba-ViTrack [44] used lightweight ViTs and an adaptive, background-aware token computation method to enhance efficiency for real-time UAV tracking. However, the variable token number in this approach necessitates unstructured access operations, leading to significant time costs. In this work, we aim to improve the efficiency of ViTs for UAV tracking through knowledge distillation, a more structured method.

2.2. Occlusion-Robust Feature Representation.

Occlusion-robust feature representation is crucial in computer vision and image processing. It involves developing methods that can recognize and process objects in images even when parts are hidden or occluded [62, 76]. Early efforts often relied on handcrafted features, active appearance models, motion analysis, sensor fusion, etc [7, 33, 51, 71]. While effective in some cases, these methods struggled with the complexity and variability of real-world visual data. The advent of deep learning revolutionized the field. Many studies have applied Convolutional Neural Networks (CNNs) and other deep architectures to

extract occlusion-robust representations [35, 62, 66, 76]. These approaches use deep models to capture complex patterns and variations in visual data, making learned features resilient to occlusions and having proven valuable for many computer vision applications, such as action recognition [17, 88], pose estimation [62, 95], and object detection [12, 36]. The exploration of occlusion-robust representations in visual tracking has also demonstrated great success [1, 6, 27, 34, 39, 58, 59, 61, 94]. However, to our knowledge, there is a dearth of research to explore learning occlusion-robust ViTs particularly in a unified framework for UAV tracking. In this study, we delve into the exploration of learning occlusion-robust feature representations based on ViTs by simulating occlusion challenges using random masking modeled by a spatial Cox process, specifically tailored for UAV tracking. This study represents the first use of ViTs for acquiring occlusion-robust feature representations in UAV tracking.

2.3. Knowledge Distillation.

Knowledge distillation is a technique used to compress models by transferring knowledge from a complex "teacher" model to a simpler "student" model, with the aim of maintaining performance while reducing computational resources and memory usage [63, 75]. It involves various types of knowledge, distillation strategies, and teacher-student architectures, typically falling into three categories: response-based, feature-based, and relation-based distillation [26, 63, 78]. Widely applied in tasks such as image classification [64], object detection [9], and neural machine translation [**?**], it offers potential to improve the efficiency and even effectiveness of deep learning models. Recently, it has been successfully utilized to enhance the efficiency of DL-based trackers. For instance, Li et al. [42] used mask-guided self-distillation to compress Siamese-based visual trackers. Sun et al. [72] introduced a lightweight dual Siamese tracker for hyperspectral object tracking, using a spatial-spectral knowledge distillation method to learn from a deep tracker. However, these techniques are mainly Siamese-based and tailored to specific tracking frameworks, posing challenges for adaptation to our ViT-based approach. In this study, we propose a simple yet effective feature-based knowledge distillation method, in which the student adaptively replicate the behavior of the teacher based on the complexity of the tracking task during training.

3. Method

In this section, we first provide a brief overview of our end-to-end tracking framework, named ORTrack, as shown in Figure 2. Then, we introduce the occlusion-robust representation learning based on spatial Cox processes and the method of adaptive knowledge distillation. Finally, we detail the prediction head and training loss.

3.1. Overview

The proposed ORTrack introduces an novel single-stream tracking framework, featuring a spatial Cox process-based masking for occlusion-robust representation learning and an adaptive feature-based knowledge distillation pipeline. ORTrack consists of two sequential training phases: the teacher model training pipeline for learning occlusion-robust representations, followed by the student training pipeline involving adaptive knowledge distillation. In the teacher model training phase, the input includes a target template $Z \in \mathbb{R}^{3\times H_z\times W_z}$ of spatial size $H_z \times W_z$, a randomly masked target template $Z' = \mathfrak{m}(Z)$, and a search image $X \in \mathbb{R}^{3\times H_x\times W_x}$ of spatial size $H_x \times W_x$, where $\mathfrak{m}(\cdot)$ represents the random masking operation that masks out non-overlap patches of size $b \times b$ with a certain masking ratio σ. To achieve occlusion-robust representation with ViTs, we minimize the mean squared error (MSE) between two versions of the template representation: one with random masking and one without. During the training of the student model, the teacher's weights remain fixed while both the teacher and student models receive inputs Z and X. Let $\mathfrak{B}_T$ and $\mathfrak{B}_S$ represent the backbones of the teacher and student, respectively. In our implementation, $\mathfrak{B}_T$ and $\mathfrak{B}_S$ share the same structure of the ViT layer but differ in the number of layers. Feature-based knowledge distillation is used to transfer the knowledge embedded in the teacher model's backbone features to the student model through an adaptive distillation loss.

3.2. Occlusion-Robust Representations (ORR) Based on Spatial Cox Processes

To begin, we describe two random masking operations used to simulate occlusion challenges: one from MAE [28] and our proposed method based on a Spatial Cox process, denoted by $\mathfrak{m}_{\mathrm{U}}$ and $\mathfrak{m}_{\mathrm{C}}$, respectively. Although $\mathfrak{m}_{\mathrm{U}}$ allows the model to learn robust representations that are less sensitive to noise or missing information by randomly ignoring certain parts of the input data during training [28], it is less effective when used to simulate occlusion since each spatial position (in the sense of block size) is masked out with equal probability, especially in our situation where the target template generally contains background. To ensure that the target is masked out as expected with higher probabilities at a given masking ratio, thereby making the occlusion simulation more effective, we employ a finite Cox process [32] to model this masking operation, which is detailed as follows.

Define two associated random matrices $\mathbf{m} = (m_{i,j})$, $\mathbf{b} = (b_{i,j})$, $1 \leqslant i \leqslant H_z/b, 1 \leqslant j \leqslant W_z/b$, where $m_{i,j} \backsim \mathcal{U}(0,1)$ (i.e., $m_{i,j}$ follows a uniform distribution over the interval [0, 1]), $b_{i,j} \in \{0,1\}$ equals 1 if $m_{i,j} \in \mathrm{TopK}(\mathbf{m}, K)$, and 0 otherwise. $\mathrm{TopK}(\mathbf{m}, K)$ returns the $K = \lfloor (1-\sigma)H_zW_z \rceil$ largest elements from $\mathbf{m}$, where $\lfloor x \rceil$ rounds x to

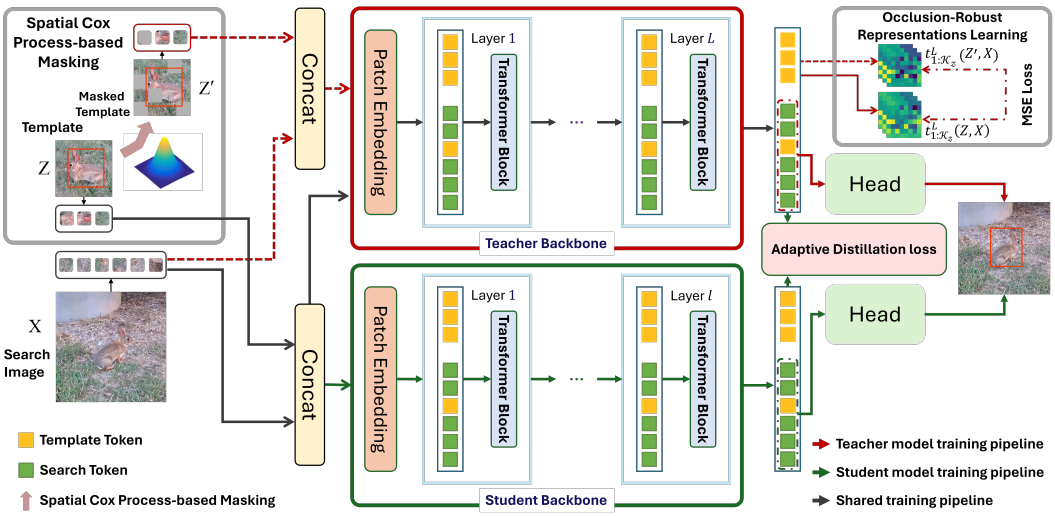


Figure 2. Overview of the proposed ORTrack framework, which includes separate training pipelines for a teacher and a student model. Note that the spatial Cox process-based masking and occlusion-robust representation learning are applied only in the teacher pipeline. Once the teacher is trained, its weights are fixed for training the student model with the proposed adaptive knowledge distillation.

the nearest integer. Mathematically, $\mathfrak{m}_{\mathrm{U}}(Z) = Z \odot (\mathbf{b} \otimes \mathbf{1})$, where $\odot$ denotes the Hadamard product and $\otimes$ denotes the tensor product, $\mathbf{1}$ is an all-ones matrix of size $b \times b$. Before defining $\mathfrak{m}_{\mathrm{C}}$, we establish core notations relevant to spatial Cox processes. It extend the concept of spatial inhomogeneous Poisson point processes by incorporating a random intensity function, which, in turn, is defined as a Poisson point process with an intensity determined by a location-dependent function in the underlying space. For Euclidean space $\mathbb{R}^2$, an inhomogeneous Poisson point process is defined by a locally integrable positive intensity function $\lambda\colon \mathbb{R}^2 \to [0, \infty)$, such that for every bounded region $\mathcal{B}$ the integral $\Lambda(\mathcal{B}) = \int_{\mathcal{B}} \lambda(x, y)\, \mathrm{d}x\mathrm{d}y$ is finite, where $\Lambda(\mathcal{B})$ has the interpretation of being the expected number of points of the Poisson process located in $\mathcal{B}$, and for every collection of disjoint bounded Borel measurable sets $\mathcal{B}_1, ..., \mathcal{B}_k$ [60], its number distributions is defined by $\Pr\{\mathrm{N}(\mathcal{B}_i) = n_i, i = 1, \ldots, k\} = \prod_{i=1}^{k} \frac{(\Lambda(\mathcal{B}_i))^{n_i}}{n_i!} e^{-\Lambda(\mathcal{B}_i)}$, $n_i \in \mathbb{Z}^{0+}$, where Pr denotes the probability measure, N indicates the random counting measure such that $\Lambda(\mathcal{B}) = \mathbb{E}[\mathrm{N}(\mathcal{B})]$, $\mathbb{E}$ is the expectation operator. In particular, the conditional distribution of the points in a bounded set $\mathcal{B}$ given that $\mathrm{N}(\mathcal{B}) = n \in \mathbb{Z}^{0+}$ is not uniform, and $f_n(p_1, ..., p_n) = \prod_{n}^{i=1} \frac{\lambda(p_i)}{\Lambda(\mathcal{B})}$, $p_1, ..., p_n \in \mathcal{B}$ defines the corresponding location density function of the n points. Since a Cox process can be regarded as the result of a two-stage random mechanism for which it is sometimes termed 'doubly stochastic Poisson process' [32], the finite Cox processes can be simulated in a straightforward way based on the hierarchical nature of the model. Specifically, in the first step, the intensity $\lambda(x, y)$ is generated. In the second step, an inhomogeneous Poisson point process is simulated using the generated $\lambda(x, y)$ [32, 53]. The thinning algorithm [11] is used here for simulating inhomogeneous Poisson point processes. It involves simulating a homogeneous Poisson point process with a higher rate than the maximum possible rate of the inhomogeneous process, and then "thinning" out the generated points to match the desired intensity function.

In this work, the randomness of the intensity function is modeled by a random variable Γ that has a Poisson distribution with expectation of ς, namely, $\Pr\{\Gamma = k\} = \frac{\varsigma^k e^{-\varsigma}}{k!}$, where $k \in \mathbb{Z}^{0+}$. The intensity function of the inhomogeneous Poisson point process is then given by

$$\lambda(x, y) = \frac{\Gamma e^{-(x^2+y^2)}}{\int_{\mathcal{B}} e^{-(x^2+y^2)} dx dy}. \tag{1}$$

Note that $\lambda(x, y)$ is a bell-shape function that gives more intensities to the central area of $\mathcal{B}$. Let $\mathcal{B}$ denote the rectangle region of size $H_z/b \times W_z/b$ representing the template region. If we simulate the Cox process within $\mathcal{B}$ and denote a resulted point pattern by Ξ, we can obtain a matrix $\mathbf{b}' = (b'_{i,j})_{1 \leqslant i \leqslant H_z/b, 1 \leqslant i \leqslant W_z/b}$, where $b'_{i,j}$ equals 1 if $(i, j) \in \Xi$, and 0 otherwise, with which our $\mathfrak{m}_{\mathrm{C}}$ can be defined as $\mathfrak{m}_{\mathrm{C}}(Z) = Z \odot (\mathbf{b}' \otimes \mathbf{1})$. It is worthy of note that if $\varsigma = \lfloor (1 - \sigma) H_z W_z \rceil$, since $\mathbb{E}[\Lambda(\mathcal{B})] = \mathbb{E}[\int_{\mathcal{B}} \lambda(x, y) dx dy] = \mathbb{E}[\Gamma] = \varsigma$, in this case, the expected masking ratio of our masking operation is equal to the masking ratio of $\mathfrak{m}_{\mathrm{C}}$. Thus, in addition to inhomogeneous intensity, our method can simulate more diverse pattern of occlusion due to the introduced randomness of the masking ratio.

We denote the total number of tokens by $\mathcal{K}$, the embedding dimension of each token by d, and all the tokens out-

put by the L-th layer of $\mathfrak{B}_T$ with respect to inputs X and Z by $\mathbf{t}^L_{1:\mathcal{K}}(Z, X; \mathfrak{B}_T) \in \mathbb{R}^{\mathcal{K}\times d}$. Let $\mathbf{t}^L_{\mathcal{K}_Z\cup\mathcal{K}_X}(Z, X; \mathfrak{B}_T) = \mathbf{t}^L_{1:\mathcal{K}}(Z, X; \mathfrak{B}_T)$, where $\mathcal{K}_Z \cup \mathcal{K}_X = [1, \mathcal{K}]$, $\mathbf{t}^L_{\mathcal{K}_Z}$ and $\mathbf{t}^L_{\mathcal{K}_X}$ represent the tokens corresponding to the template and the search image, respectively. By the same token, the output tokens corresponding to inputs X and Z' are $\mathbf{t}^L_{1:\mathcal{K}}(Z', X; \mathfrak{B}_T)$. The feature representations of Z and Z' can be recovered by tracking their token indices in respective ordered sequences, which specifically are $t^L_{1:\mathcal{K}_z}(Z, X; \mathfrak{B}_T)$ and $t^L_{1:\mathcal{K}_z}(Z', X; \mathfrak{B}_T)$, respectively. The core idea of our occlusion-robust representations learning is that the mean square error between the feature representation of Z and that of Z' is minimized, which is implemented by minimizing the following MSE loss,

$$\mathcal{L}_{orr} = ||t^L_{1:\mathcal{K}_z}(Z, X; \mathfrak{B}_T) - t^L_{1:\mathcal{K}_z}(Z', X; \mathfrak{B}_T)||^2. \quad (2)$$

During inference, only $[Z, X]$ is input to the model without the need for random template masking. Consequently, our method incurs no additional computational cost during inference. Notably, our method is independent of the ViTs used, any efficient ViTs can work in our framework.

3.3. Adaptive Feature-Based Knowledge Distillation (AFKD)

Feature-based knowledge distillation is a technique in machine learning that trains a smaller student model to mimic a larger teacher model, which, instead of focusing only on final outputs, transfers intermediate features or representations from the teacher to the student [26, 78]. This method uses the detailed internal representations from the teacher model to improve the student's learning process. However, there is a risk that the student model might overfit to the specific features of the teacher model, rather than generalizing well to new data. This can be particularly problematic if the teacher model has learned spurious correlations in the data. To combat this, we propose adaptively transferring knowledge based on the difficulty of the tracking task. We quantify this difficulty using the deviation of the GIoU loss [67] (see Section 3.4) from its average value, calculated between the student's prediction and the ground truth. Adapting knowledge transfer based on difficulty ensures that the student model doesn't heavily adjust its weights on easy tasks, which it can handle already probably due to its generalizability. Instead, it focuses more on challenging scenarios where its feature representation is less effective.

Additionally, the choice of teacher-student architectures is crucial in knowledge distillation. Given the wide array of possible student models, we adopt a self-similar approach where the student model mirrors the teacher's architecture but employs a smaller ViT backbone, using fewer ViT blocks. This strategy simplifies the design and eliminates the need for additional alignment techniques that would otherwise be necessary due to mismatched feature dimensions. Lastly, layer selection and the metric of feature similarity are also crucial aspects of feature-based knowledge distillation. Given MSE's popularity in feature-based knowledge distillation and to avoid potential complexity associated with using multiple layers, we employ MSE to penalize differences between the output feature representations of both the teacher and student model's backbones, i.e., $t^L_{1:\mathcal{K}}(Z, X; \mathfrak{B}_T)$ and $t^L_{1:\mathcal{K}}(Z, X; \mathfrak{B}_S)$. The proposed adaptive knowledge distillation loss is defined by

$$\mathcal{L}_{afkd} = (\alpha+\beta(\mathcal{L}_{iou}-\overline{\mathcal{L}_{iou}}))||t^L_{1:\mathcal{K}}(Z, X; \mathfrak{B}_T)-t^L_{1:\mathcal{K}}(Z, X; \mathfrak{B}_S)||^2, \quad (3)$$

where $\alpha+\beta(\mathcal{L}_{iou}-\overline{\mathcal{L}_{iou}}) := \varpi(\mathcal{L}_{iou}; \alpha, \beta)$ is a function of the deviation of GIoU loss from its average, with slop α and intercept β, used to quantify the difficulty of the tracking task.

3.4. Prediction Head and Training Loss

Following the corner detection head in [13, 91], we use a prediction head consisting of multiple Conv-BN-ReLU layers to directly estimate the bounding box of the target. The output tokens corresponding to the search image are first reinterpreted to a 2D spatial feature map and then fed into the prediction head. The head outputs a local offset $\mathbf{o} \in [0, 1]^{2\times H_x/P\times W_x/P}$, a normalized bounding box size $\mathbf{s} \in [0, 1]^{2\times H_x/P\times W_x/P}$, and a target classification score $\mathbf{p} \in [0, 1]^{H_x/P\times W_x/P}$ as prediction outcomes. The initial estimation of the target position depends on identifying the location with the highest classification score, i.e., $(x_c, y_c) = \text{argmax}_{(x,y)}\mathbf{p}(x, y)$. The final target bounding box is estimated by $\{(x_t, y_t); (w, h)\} = \{(x_c, y_c) + \mathbf{o}(x_c, y_c); \mathbf{s}(x_c, y_c)\}$. For the tracking task, we adopt the weighted focal loss [40] for classification, a combination of L_1 loss and GIoU loss [67] for bounding box regression. The total loss for tracking prediction is:

$$\mathcal{L}_{pred} = \mathcal{L}_{cls} + \lambda_{iou}\mathcal{L}_{iou} + \lambda_{L_1}\mathcal{L}_{L_1}, \quad (4)$$

where the constants λ_{iou} = 2 and λ_{L_1}= 5 are set as in [13, 91]. The overall loss $\mathcal{L}_T = \mathcal{L}_{pred} + \gamma\mathcal{L}_{orr}$ is used to train the teacher end-to-end after loading the pretrained weights of the ViT trained with ImageNet [68], where the constant γ is set to 2.0×10^{-4}. After this training, we fix the weights of the teacher model, and employ the overall loss $\mathcal{L}_S = \mathcal{L}_{pred} + \mathcal{L}_{afkd}$, for end-to-end knowledge distillation training.

4. Experiments

We evaluate our method on four UAV tracking benchmarks: DTB70 [45], UAVDT [19], VisDrone2018 [98], and UAV123 [57]. All experiments run on a PC with an i9-10850K processor, 16GB RAM, and an NVIDIA TitanX GPU. We compare our method against 26 state-of-the-art trackers, using their official codes and hyper-

Table 1. Precision (Prec.), success rate (Succ.), and speed (FPS) comparison between ORTrack and lightweight trackers on four UAV tracking benchmarks, i.e., DTB70 [45], UAVDT [19], VisDrone2018 [98], and UAV123 [57]. Red, blue and green indicate the first, second and third place. Note that the percent symbol (%) is omitted for all Prec. and Succ. values.

	Method	Source	DTB70		UAVDT		VisDrone2018		UAV123		Avg.		Avg.FPS		FLOPs	Param.
			Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	Prec.	Succ.	GPU	CPU	(GMac)	(M)
DCF-based	KCF [29]	TAPMI 15	46.8	28.0	57.1	29.0	68.5	41.3	52.3	33.1	56.2	32.9	-	**624.3**	-	-
	fDSST [16]	TPAMI 17	53.4	35.7	66.6	38.3	69.8	51.0	58.3	40.5	62.0	41.4	-	**193.4**	-	-
	ECO_HC [14]	CVPR 17	63.5	44.8	69.4	41.6	80.8	58.1	71.0	49.6	71.2	48.5	-	**83.5**	-	-
	AutoTrack [46]	CVPR 20	71.6	47.8	71.8	45.0	78.8	57.3	68.9	47.2	72.8	49.3	-	57.8	-	-
	RACF [41]	PR 22	72.6	50.5	77.3	49.4	83.4	60.0	70.2	47.7	75.9	51.8	-	35.6	-	-
CNN-based	HiFT [4]	ICCV 21	80.2	59.4	65.2	47.5	71.9	52.6	78.7	59.0	74.0	54.6	160.3	-	7.2	9.9
	TCTrack [5]	CVPR 22	81.2	62.2	72.5	53.0	79.9	59.4	80.0	60.5	78.4	58.8	149.6	-	8.8	9.7
	SGDViT [90]	ICRA 23	78.5	60.4	65.7	48.0	72.1	52.1	75.4	57.5	72.9	54.5	110.5	-	11.3	23.3
	DRCI [93]	ICME 23	81.4	61.8	**84.0**	59.0	83.4	60.0	76.7	59.7	81.4	60.1	**281.3**	**62.7**	3.6	8.8
	PRL-Track [22]	IROS 24	79.5	60.6	73.1	53.5	72.6	53.8	79.1	59.3	76.1	56.8	132.3	-	7.4	12.0
ViT-based	Aba-ViTrack [44]	ICCV 23	**85.9**	**66.4**	**83.4**	**59.9**	**86.1**	**65.3**	**86.4**	**66.4**	**85.5**	**64.5**	181.5	50.3	2.4	8.0
	SMAT [25]	WACV 24	81.9	63.8	80.8	58.7	82.5	63.4	81.8	64.6	81.8	62.6	126.8	-	3.2	8.6
	AVTrack-DeiT [47]	ICML 24	**84.3**	**65.0**	82.1	58.7	**86.0**	**65.3**	**84.8**	**66.8**	**84.2**	**63.8**	**260.3**	**59.8**	0.97-1.9	3.5-7.9
	ORTrack-DeiT	**Ours**	**86.2**	**66.4**	**83.4**	**60.1**	**88.6**	**66.8**	**84.3**	**66.4**	**85.6**	**65.0**	226.4	55.4	2.4	7.9
	ORTrack-D-DeiT		83.7	**65.1**	**82.5**	**59.7**	84.6	**63.9**	84.0	**66.1**	83.7	63.7	**292.3**	**64.7**	1.5	5.3

parameters. We evaluate our approach against 13 state-of-the-art (SOTA) lightweight trackers (see Table 1) and 14 SOTA deep trackers designed specifically for generic visual tracking (refer to Table 2).

4.1. Implementation Details

We adopt different ViTs as backbones, including ViT-tiny [18], Eva-tiny [21], and DeiT-tiny [73], to build three trackers for evaluation: ORTrack-ViT, ORTrack-Eva, and ORTrack-DeiT. The head of ORTrack consists of a stack of four Conv-BN-ReLU layers. The search region and template sizes are set to 256×256 and 128×128, respectively. A combination of training sets from GOT-10k [30], LaSOT [20], COCO [48], and TrackingNet [56] is used for the training. The batch size is set to 32. We employ the AdamW optimizer [50], with a weight decay of 10^{-4} and an initial learning rate of 4×10^{-5}. The training is conducted over 300 epochs, with 60,000 image pairs processed in each epoch. The learning rate is reduced by a factor of 10 after 240 epochs.

4.2. State-of-the-art Comparison

Comparison with Lightweight Trackers. The overall performance of our ORTrack in comparison to 13 competing trackers on the four benchmarks is displayed in Table 1. As can be seen, our trackers demonstrate superior performance among all these trackers in terms of average (Avg.) precision (Prec.), success rate (Succ.) and speeds. On average, RACF [41] demonstrated the highest Prec. (75.9%) and Succ. (51.8%) among DCF-based trackers, DRCI [93] achieves the highest precision and success rates, with 81.4% and 60.1%, respectively, among CNN-based trackers. However, the average Prec. and Succ. of all our trackers are greater than 82.0% and 62.0%, respectively, clearly surpassing DCF- and CNN- based approaches. Additionally, our ORTrack-DeiT achieves the highest Avg. Prec. and Avg. Succ. of 85.6% and 65.0%, respectively, among all competing trackers. Although Aba-ViTrack achieves performance close to our ORTrack-DeiT, its GPU speed is significantly lower, with a 23.6% relative gap. Notably, when the proposed adaptive knowledge distillation is applied to ORTrack-DeiT, the resulting student model, ORTrack-D-DeiT, shows a significant speed increase: 29.1% on GPU and 16.8% on CPU. This improvement is accompanied by a minimal reduction in accuracy, with only a 1.9% decrease in Avg. Prec. and a 1.3% decrease in Avg. Succ.. All proposed trackers can run in real-time on a single CPU*, and our ORTrack-DeiT sets a new performance record for real-time UAV tracking. We also compare the floating point operations per second (FLOPs) and number of parameters (Params.) of our method with CNN-based and ViT-based trackers in Table 1. Our method demonstrates a relatively lower parameter count and reduced computational complexity compared to these approaches. Notably, since AVTrack-DeiT tracker features adaptive architectures, the FLOPs and parameters range from minimum to maximum values. These results highlight our method's effectiveness and its state-of-the-art performance.

Comparison with Deep Trackers. The proposed ORTrack-DeiT is also compared with 14 SOTA deep trackers in Table 2, which shows precision (Prec.) and GPU speed on VisDrone2018. Our ORTrack-DeiT surpasses all other methods in both metrics, demonstrating its superior accuracy and speed. Although trackers like AQATrack [87], HIPTrack [2], and ROMTrack [3] achieve precision comparable to our ORTrack-DeiT, their GPU speeds are much slower. Specifically, our method is 4, 6, and 4 times faster than AQATrack, HIPTrack, and ROMTrack, respectively.

Attribute-Based Evaluation. To access our method's robustness against target occlusion, we compare ORTrack-

*Real-time performance applies to platforms similar to or more advanced than ours.

Table 2. Precision (Prec.) and speed (FPS) comparison between ORTrack-DeiT and deep-based trackers on VisDrone2018 [98].

Tracker	Source	Prec.	Succ.	FPS	Tracker	Source	Prec.	Succ.	FPS	Tracker	Source	Prec.	Succ.	FPS
ORTrack-DeiT	**Ours**	**88.6**	**66.8**	**206.2**	ZoomTrack [38]	NIPS 23	81.4	63.4	61.7	SimTrack [8]	ECCV 22	80.0	60.9	**69.7**
AQATrack [87]	CVPR 24	**87.2**	**66.9**	53.4	SeqTrack [10]	CVPR 23	85.3	65.8	15.3	ToMP [55]	CVPR 22	84.1	64.4	21.4
HIPTrack [2]	CVPR 24	**86.7**	**67.1**	31.3	MAT [96]	CVPR 23	81.6	62.2	**68.4**	KeepTrack [54]	ICCV 21	84.0	63.5	20.3
EVPTrack [69]	AAAI 24	84.5	65.8	22.1	SparseTT [23]	IJCAI 22	81.4	62.1	30.2	SAOT [97]	ICCV 21	76.9	59.1	35.4
ROMTrack [3]	ICCV 23	86.4	66.7	51.1	OSTrack [91]	ECCV 22	84.2	64.8	62.7	PrDiMP50 [15]	CVPR 20	79.4	59.7	42.6

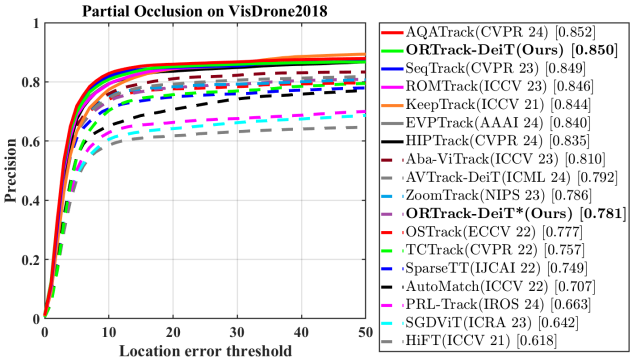


Figure 3. Attribute-based comparison on the partial occlusion subset of VisDrone2018 [98]. ORTrack-DeiT* refers to ORTrack-DeiT without applying the occlusion-robust enhancement.

DeiT alongside 16 SOTA trackers on the partial occlusion subset of VisDrone2018. Additionally, we also assess the baseline ORTrack-DeiT*, i.e., ORTrack-DeiT without applying the proposed method for learning Occlusion-Robust Representation (ORR), for comparison. The precision plot are presented in Fig. 3, with additional attribute-based evaluation results provided in the supplemental materials. As observed, ORTrack-DeiT achieves the second-highest precision (85.0%), just slightly behind the first-ranked tracker AQATrack by 0.2%. Remarkably, incorporating the proposed components leads to a significant improvement over ORTrack-DeiT*, with increases of 6.9% in Prec., well underscoring the effectiveness of our method.

4.3. Ablation Study

Table 3. Effect of ORR and AFKD on the baseline trackers.

Tracker	ORR	AFKD	UAVDT Prec.	UAVDT Succ.	FPS
ORTrack-ViT			77.0	55.6	216.2
	✓		**80.3**$_{\uparrow 3.3}$	**58.2**$_{\uparrow 2.6}$	-
	✓	✓	79.1$_{\uparrow 2.1}$	57.5$_{\uparrow 1.9}$	**290.3**$_{\uparrow 34\%}$
ORTrack-Eva			78.1	56.6	238.3
	✓		**80.8**$_{\uparrow 2.7}$	**58.7**$_{\uparrow 2.1}$	-
	✓	✓	79.5$_{\uparrow 1.4}$	57.8$_{\uparrow 1.2}$	**308.8**$_{\uparrow 30\%}$
ORTrack-DeiT			78.6	56.7	218.4
	✓		**83.4**$_{\uparrow 4.8}$	**60.1**$_{\uparrow 3.4}$	-
	✓	✓	82.5$_{\uparrow 3.9}$	59.7$_{\uparrow 3.0}$	**298.7**$_{\uparrow 36\%}$

Effect of Occlusion-Robust Representations (ORR) and Adaptive Feature-Based Knowledge Distillation (AFKD). To demonstrate the effectiveness of the proposed ORR and AFKD, Table 3 shows the evaluation results on UAVDT dataset as these components are gradually integrated into the baselines. To avoid potential variations due to randomness, we only present the speed of the baseline, since the GPU speeds of the baseline and its ORR-enhanced version are theoretically identical. As can bee seen, the incorporation of ORR significantly enhances both Prec. and Succ. for all baseline trackers. Specifically, the Prec. increases for ORTrack-ViT, ORTrack-Eva, and ORTrack-DeiT are 3.3%, 2.7%, and 4.8%, respectively, while the Succ. increases are 2.6%, 2.1%, and 3.1%, respectively. These significant enhancements highlight the effectiveness of ORR in improving tracking precision. The further integration of AFKD results in consistent improvements in GPU speeds, with only slight reductions in Prec. and Succ. Specifically, all baseline trackers experience GPU speed enhancements of over 30.0%, with ORTrack-DeiT showing an impressive 36.0% improvement. These results affirm the effectiveness of AFKD in optimizing tracking efficiency while maintaining high tracking performance.

Table 4. Impact of various Masking Operators on performance.

Method	$\mathfrak{m}_U$	$\mathfrak{m}_C$	SAM[37]	AdAutoMix[65]	CutMix[92]	VisDrone2018 Prec.	VisDrone2018 Succ.
ORTrack-DeiT						81.6	62.2
	✓					86.7	65.4
		✓				**88.6**	**66.8**
			✓			86.8	65.6
				✓		84.3	63.8
					✓	85.7	64.2

Effect of Masking Operators. To demonstrate the superiority of the proposed masking operator in terms of performance, we evaluate ORTrack-DeiT with various implementations of masking operators (i.e., $\mathfrak{m}_U$, $\mathfrak{m}_C$, and SAM [37]) alongside data mixing augmentation methods (i.e., AdAutoMix [65] and CutMix [92]). The evaluation results on VisDrone2018 are presented in Table 4. As shown, although using SAM, AdAutoMix, and CutMix improves performance, the best result achieved with SAM is only comparable to the performance of our $\mathfrak{m}_U$ masking operator. When $\mathfrak{m}_C$ is applied, the improvements are even more substantial, with increases of 7.0% and 4.6%, respectively. These results validate the effectiveness of the proposed ORR component and particularly demonstrate the superiority of the masking operator based on spatial Cox processes.

Table 5. Impact of the adaptive knowledge distillation loss on the generalizability on LaSOT and TrackingNet.

Method	KD	AFKD	LaSOT			TrackingNet		
			AUC	Pnorm	P	AUC	Pnorm	P
ORTrack-DeiT			53.7	60.8	52.6	72.8	77.8	67.1
	✓		54.0	61.2	53.2	73.1	78.4	67.4
		✓	**54.6**	**62.6**	**54.3**	**73.7**	**79.1**	**68.2**

Impact of the Adaptive Knowledge Distillation Loss. To assess the impact of the adaptive knowledge distillation loss on generalizability, we train ORTrack-DeiT using GOT-10K with $\varpi(\mathcal{L}_{iou};\alpha,\beta)$ and $\varpi(\mathcal{L}_{iou};\alpha,0)$ separately, then evaluate them on LaSOT and TrackingNet. The results are shown in Table 5. Note that $\varpi(\mathcal{L}_{iou};\alpha,0)$ degenerates to a non-adaptive knowledge distillation loss as it becomes a constant. As can be seen, AFKD demonstrates greater performance improvements than KD. For instance, using AFKD results in additional gains of over 1.1% in P_{norm} and P on LaSOT, demonstrating its superior generalizability.

Table 6. Application of our ORR component to three SOTA trackers: ARTrack [81], GRM [24], and DropTrack[82].

Tracker	ORR	UAVDT		VisDrone2018	
		Prec.	Succ.	Prec.	Succ.
ARTrack[81]		77.1	54.6	77.7	59.5
	✓	**78.5**$_{\uparrow 1.4}$	**55.8**$_{\uparrow 1.2}$	**79.5**$_{\uparrow 1.8}$	**60.8**$_{\uparrow 1.3}$
GRM[24]		79.0	57.7	82.7	63.4
	✓	**81.7**$_{\uparrow 1.7}$	**59.3**$_{\uparrow 1.6}$	**84.8**$_{\uparrow 2.1}$	**64.6**$_{\uparrow 1.2}$
DropTrack[82]		76.9	55.9	81.5	62.7
	✓	**78.7**$_{\uparrow 1.8}$	**57.4**$_{\uparrow 1.5}$	**82.8**$_{\uparrow 1.3}$	**64.2**$_{\uparrow 1.5}$

Application to SOTA trackers. To show the wide applicability of our proposed method, we incorporate the proposed ORR into three existing SOTA trackers: ARTrack [81], GRM [24], and DropTrack [82]. Please note that we replace the model's original backbones with ViT-tiny [18] to reduce training time. As shown in Table 6, incorporating ORR results in significant improvements in both precision and success rates for the three baseline trackers. Specifically, ARTrack, GRM, and DropTrack demonstrate an improvement of more than 1.2% in both precision and success rate across two datasets. These experimental results demonstrate that the proposed ORR component can be seamlessly integrated into existing tracking frameworks, significantly improving tracking accuracy.

Qualitative Results. Several qualitative tracking results of ORTrack-DeiT and seven SOTA UAV trackers are shown in Fig. 4. As can be seen, only our tracker successfully tracks the targets in all challenging examples, where pose variations, background clusters, and scale variations are presented. In these cases, our method performs significantly better and is more visually appealing, bolstering the effectiveness of the proposed method for UAV tracking.

Figure 5 shows attention and feature maps produced by

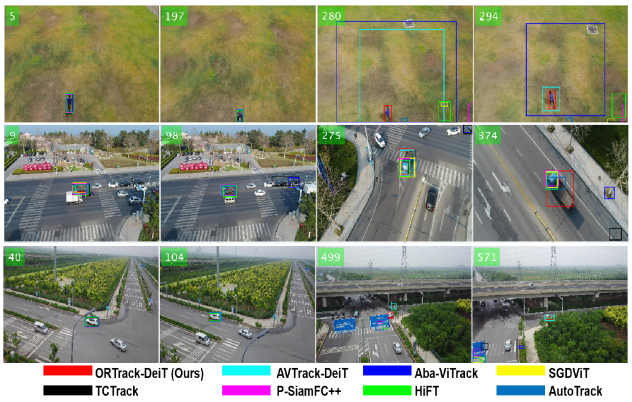


Figure 4. Qualitative evaluation on 3 video sequences from, respectively, UAV123 [57], UAVDT [19], and VisDrone2018 [98] (i.e., person9, S1607, and uav0000180_00050_s).

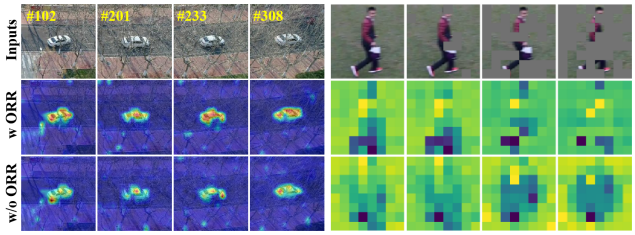


Figure 5. Visualize the attention map (left) and feature map (right) of the target images. The first row displays the search and masked images with masking ratios of 0%, 10%, 30%, and 70%. The second and third rows show the attention and feature maps generated by ORTrack-DeiT, with and without ORR, respectively.

ORTrack-DeiT, with and without occlusion-robust enhancement. We observe that ORTrack-DeiT with ORR maintains a clearer focus on the targets and exhibits more consistent feature maps across masking ratios. These results support the effectiveness of our ORR component.

5. Conclusion

In view of the common challenges posed by target occlusion in UAV tracking, in this work, we proposed to learn Occlusion-Robust Representation (ORR) by imposing an invariance of feature representation of the target with respect to random masking modeled by a spatial Cox process. Moreover, we propose an Adaptive Feature-Based Knowledge Distillation (AFKD) to enhance efficiency. Our approach is notably straightforward and can be easily integrated into other tracking frameworks. Extensive experiments across multiple UAV tracking benchmarks validate the effectiveness of our method, demonstrating that our ORTrack-DeiT achieves SOTA performance.

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References

[1] Wesam A. Askar, Osama Elmowafy, Anca L. Ralescu, Aliaa Abdel-Halim Youssif, and Gamal A. Elnashar. Occlusion detection and processing using optical flow and particle filter. *Int. J. Adv. Intell. Paradigms*, 15:63–76, 2020. 3

[2] Wenrui Cai, Qingjie Liu, and Yunhong Wang. Hiptrack: Visual tracking with historical prompts. In *CVPR*, pages 19258–19267, 2024. 6, 7

[3] Yidong Cai, Jie Liu, Jie Tang, and Gangshan Wu. Robust object modeling for visual tracking. In *ICCV*, pages 9589–9600, 2023. 6, 7

[4] Ziang Cao, Changhong Fu, Junjie Ye, Bowen Li, and Yiming Li. Hift: Hierarchical feature transformer for aerial tracking. In *ICCV*, pages 15457–15466, 2021. 1, 2, 6

[5] Ziang Cao, Ziyuan Huang, Liang Pan, Shiwei Zhang, Ziwei Liu, and Changhong Fu. Tctrack: Temporal contexts for aerial tracking. In *CVPR*, pages 14798–14808, 2022. 1, 2, 6

[6] Satyaki Chakraborty and Martial Hebert. Learning to track object position through occlusion. *ArXiv*, abs/2106.10766, 2021. 3

[7] T-H Chang and Shaogang Gong. Tracking multiple people with a multi-camera system. In *Womot*, pages 19–26, 2001. 2

[8] Boyu Chen, Peixia Li, Lei Bai, Lei Qiao, and et al. Backbone is all your need: a simplified architecture for visual object tracking. In *ECCV*, pages 375–392, 2022. 1, 7

[9] Guobin Chen, Wongun Choi, Xiang Yu, Tony Han, and et al. Learning efficient object detection models with knowledge distillation. *NIPS*, 30, 2017. 3

[10] Xin Chen, Houwen Peng, Dong Wang, Huchuan Lu, and Han Hu. Seqtrack: Sequence to sequence learning for visual object tracking. In *CVPR*, pages 14572–14581, 2023. 7

[11] Yuanda Chen. Thinning algorithms for simulating point processes. *Florida State University, Tallahassee, FL*, 2016. 4

[12] Cheng Chi, Shifeng Zhang, Junliang Xing, Zhen Lei, S. Li, and Xudong Zou. Pedhunter: Occlusion robust pedestrian detector in crowded scenes. *ArXiv*, abs/1909.06826, 2019. 3

[13] Yutao Cui, Cheng Jiang, and et al. Mixformer: End-to-end tracking with iterative mixed attention. In *CVPR*, pages 13608–13618, 2022. 1, 2, 5

[14] Martin Danelljan, Goutam Bhat, Fahad Shahbaz Khan, and Michael Felsberg. Eco: Efficient convolution operators for tracking. In *CVPR*, pages 6638–6646, 2017. 6

[15] Martin Danelljan, Luc Van Gool, and Radu Timofte. Probabilistic regression for visual tracking. In *CVPR*, pages 7181–7190, 2020. 7

[16] Martin Danelljan, Gustav Hager, Fahad Shahbaz Khan, and et al. Discriminative scale space tracking. *IEEE TPAMI*, 39(8):1561–1575, 2017. 6

[17] Soumen Das, Saroj K. Biswas, and Biswajit Purkayastha. Occlusion robust sign language recognition system for indian sign language using cnn and pose features. *Multimed. Tools. Appl*, 2024. 3

[18] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, and et al. An image is worth 16x16 words: Transformers for image recognition at scale. *ArXiv*, abs/2010.11929, 2020. 6, 8

[19] Dawei Du, Yuankai Qi, Hongyang Yu, and et al. The unmanned aerial vehicle benchmark: Object detection and tracking. In *ECCV*, pages 375–391, 2018. 5, 6, 8

[20] Heng Fan, Liting Lin, Fan Yang, and et al. Lasot: A high-quality benchmark for large-scale single object tracking. In *CVPR*, pages 5369–5378, 2018. 6

[21] Yuxin Fang, Quan Sun, Xinggang Wang, and et al. Eva-02: A visual representation for neon genesis. *Image and Vision Computing*, 149:105171, 2024. 6

[22] Changhong Fu, Xiang Lei, and et al. Progressive representation learning for real-time uav tracking. In *IROS*, pages 5072–5079, 2024. 6

[23] Zhihong Fu, Zehua Fu, Qingjie Liu, Wenrui Cai, and Yunhong Wang. Sparsett: Visual tracking with sparse transformers. *arXiv e-prints*, 2022. 7

[24] Shenyuan Gao, Chunluan Zhou, and Jun Zhang. Generalized relation modeling for transformer tracking. In *CVPR*, pages 18686–18695, 2023. 8

[25] Goutam Yelluru Gopal and Maria A Amer. Separable self and mixed attention transformers for efficient object tracking. In *WACV*, pages 6708–6717, 2024. 6

[26] Jianping Gou, Baosheng Yu, Stephen J Maybank, and Dacheng Tao. Knowledge distillation: A survey. *IJCV*, 129(6):1789–1819, 2021. 3, 5

[27] Karthik Hariharakrishnan and Dan Schonfeld. Fast object tracking using adaptive block matching. *IEEE TMM*, 7:853–859, 2005. 3

[28] Kaiming He, Xinlei Chen, Saining Xie, and et al. Masked autoencoders are scalable vision learners. In *CVPR*, pages 15979–15988, 2021. 3

[29] João F. Henriques, Rui Caseiro, Pedro Martins, and et al. High-speed tracking with kernelized correlation filters. *IEEE TPAMI*, 37:583–596, 2015. 6

[30] L. Huang, X. Zhao, and K. Huang. Got-10k: A large high-diversity benchmark for generic object tracking in the wild. *IEEE TPAMI*, (5), 2021. 6

[31] Ziyuan Huang, Changhong Fu, and et al. Learning aberrance repressed correlation filters for real-time uav tracking. In *ICCV*, pages 2891–2900, 2019. 2

[32] Janine Illian, Antti Penttinen, Helga Stoyan, and Dietrich Stoyan. *Statistical analysis and modelling of spatial point patterns*. John Wiley & Sons, 2008. 3, 4

[33] Michal Irani and Shmuel Peleg. Motion analysis for image enhancement: Resolution, occlusion, and transparency. *JVCIR*, 4(4):324–335, 1993. 2

[34] Dippal Israni and Hiren K. Mewada. Feature descriptor based identity retention and tracking of players under intense occlusion in soccer videos. *Int. J. Intell. Eng. Syst*, 2018. 3

[35] Minyang Jiang and et al. Occlusion-robust fau recognition by mining latent space of masked autoencoders. *Neurocomputing*, 569:127107, 2024. 3

[36] Jung Uk Kim, Ju Won Kwon, and et al. Bbc net: Bounding-box critic network for occlusion-robust object detection. *IEEE TCSVT*, 30:1037–1050, 2020. 3

[37] Alexander Kirillov, Eric Mintun, and et al. Segment anything. In *ICCV*, pages 4015–4026, 2023. 7

[38] Yutong Kou, Jin Gao, Bing Li, and et al. Zoomtrack: Target-aware non-uniform resizing for efficient visual tracking. *NIPS*, 36:50959–50977, 2023. 7

[39] Thijs P. Kuipers, Devanshu Arya, and Deepak K. Gupta. Hard occlusions in visual object tracking. In *ECCV Workshops*, 2020. 3

[40] Hei Law and Jia Deng. Cornernet: Detecting objects as paired keypoints. *IJCV*, 128:642–656, 2018. 5

[41] Bei Li, Ziyang Wang, and et al. Learning light-weight translation models from deep transformer. In *AAAI*, volume 35, pages 13217–13225, 2021. 1, 2, 6

[42] Luming Li, Chenglizhao Chen, and Xiaowei Zhang. Mask-guided self-distillation for visual tracking. In *ICME*, pages 1–6, 2022. 3

[43] Shuiwang Li, Xiangyang Yang, and et al. Learning target-aware vision transformers for real-time uav tracking. *IEEE TGRS*, 2024. 1

[44] Shuiwang Li, Yangxiang Yang, Dan Zeng, and Xucheng Wang. Adaptive and background-aware vision transformer for real-time uav tracking. In *ICCV*, pages 13943–13954, 2023. 1, 2, 6

[45] Siyi Li and D. Y. Yeung. Visual object tracking for unmanned aerial vehicles: A benchmark and new motion models. In *AAAI*, 2017. 5, 6

[46] Yiming Li, Changhong Fu, Fangqiang Ding, and et al. Autotrack: Towards high-performance visual tracking for uav with automatic spatio-temporal regularization. In *CVPR*, pages 11920–11929, 2020. 1, 2, 6

[47] Yongxin Li, Mengyuan Liu, You Wu, and et al. Learning adaptive and view-invariant vision transformer for real-time uav tracking. In *ICML*, 2024. 6

[48] Tsung Yi Lin, Michael Maire, and et al. Microsoft coco: Common objects in context. In *ECCV*, 2014. 6

[49] Mengyuan Liu, Yuelong Wang, and et al. Global filter pruning with self-attention for real-time uav tracking. In *BMVC*, page 861, 2022. 1

[50] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017. 6

[51] David G Lowe. Object recognition from local scale-invariant features. In *ICCV*, pages 1150–1157, 1999. 2

[52] Siyu Ma, Yuting Liu, and et al. Learning disentangled representation in pruning for real-time uav tracking. In *ACML*, pages 690–705, 2023. 1

[53] Torsten Mattfeldt. Stochastic geometry and its applications. *Journal of Microscopy*, 183:257–257, 1996. 4

[54] Christoph Mayer, Martin Danelljan, and et al. Learning target candidate association to keep track of what not to track. In *ICCV*, pages 13424–13434, 2021. 7

[55] Christoph Mayer, Martin Danelljan, and et al. Transforming model prediction for tracking. In *CVPR*, pages 8721–8730, 2022. 7

[56] Matthias Mueller, Adel Bibi, and et al. Trackingnet: A large-scale dataset and benchmark for object tracking in the wild. In *ECCV*, pages 300–317, 2018. 6

[57] Matthias Mueller, Neil G. Smith, and Bernard Ghanem. A benchmark and simulator for uav tracking. In *ECCV*, 2016. 5, 6, 8

[58] Hieu Tat Nguyen and Arnold W. M. Smeulders. Fast occluded object tracking by a robust appearance filter. *IEEE TPAMI*, 26:1099–1104, 2004. 3

[59] Hieu Tat Nguyen, Marcel Worring, and Rein van den Boomgaard. Occlusion robust adaptive template tracking. In *ICCV*, pages 678–683, 2001. 3

[60] Toby C. O'Neil. Geometric measure theory. 2002. 4

[61] Jiyan Pan and Bo Hu. Robust occlusion handling in object tracking. In *CVPR*, pages 1–8, 2007. 3

[62] Joo Hyun Park, Yeong Min Oh, and et al. Handoccnet: Occlusion-robust 3d hand mesh estimation network. In *CVPR*, pages 1486–1495, 2022. 2, 3

[63] Wonpyo Park and et al. Relational knowledge distillation. In *CVPR*, pages 3962–3971, 2019. 3

[64] Zhimao Peng, Zechao Li, Junge Zhang, and et al. Few-shot image recognition with knowledge transfer. In *ICCV*, pages 441–449, 2019. 3

[65] Huafeng Qin, Xin Jin, Yun Jiang, Mounim A El-Yacoubi, and Xinbo Gao. Adversarial automixup. *arXiv preprint arXiv:2312.11954*, 2023. 7

[66] Delin Qu, Yizhen Lao, and et al. Towards nonlinear-motion-aware and occlusion-robust rolling shutter correction. *ICCV*, pages 10646–10654, 2023. 3

[67] Seyed Hamid Rezatofighi, Nathan Tsoi, JunYoung Gwak, and et al. Generalized intersection over union:

A metric and a loss for bounding box regression. *CVPR*, pages 658–666, 2019. 2, 5

[68] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, and et al. Imagenet large scale visual recognition challenge. *IJCV*, 115:211 – 252, 2014. 5

[69] Liangtao Shi, Bineng Zhong, Qihua Liang, Ning Li, Shengping Zhang, and Xianxian Li. Explicit visual prompts for visual object tracking. In *AAAI*, 2024. 7

[70] Abhinav Shrivastava, Abhinav Kumar Gupta, and Ross B. Girshick. Training region-based object detectors with online hard example mining. In *CVPR*, pages 761–769, 2016. 2

[71] Markus Storer and et al. Active appearance model fitting under occlusion using fast-robust pca. In *VISAPP*, pages 129–136, 2009. 2

[72] Chen Sun and et al. Siamohot: A lightweight dual siamese network for onboard hyperspectral object tracking via joint spatial-spectral knowledge distillation. *IEEE TGRS*, 61:1–12, 2023. 3

[73] Hugo Touvron and et al. Training data-efficient image transformers & distillation through attention. In *ICML*, pages 10347–10357, 2021. 6

[74] Wenxuan Tu, Sihang Zhou, and et al. Hierarchically contrastive hard sample mining for graph self-supervised pretraining. *IEEE TNNLS*, PP, 2023. 2

[75] Frederick Tung and Greg Mori. Similarity-preserving knowledge distillation. In *ICCV*, pages 1365–1374, 2019. 3

[76] K. Wang and et al. Region attention networks for pose and occlusion robust facial expression recognition. *IEEE TIP*, 29:4057–4069, 2019. 2, 3

[77] Keze Wang and et al. Towards human-machine cooperation: Self-supervised sample mining for object detection. In *CVPR*, pages 1605–1613, 2018. 2

[78] Lin Wang and Kuk-Jin Yoon. Knowledge distillation and student-teacher learning for visual intelligence: A review and new outlooks. *IEEE TPAMI*, 44:3048–3068, 2020. 3, 5

[79] Xucheng Wang, Xiangyang Yang, and et al. Learning disentangled representation with mutual information maximization for real-time uav tracking. In *ICME*, pages 1331–1336, 2023. 1

[80] Xucheng Wang, Dan Zeng, Qijun Zhao, and Shuiwang Li. Rank-based filter pruning for real-time uav tracking. In *ICME*, pages 01–06, 2022. 1, 2

[81] Xing Wei, Yifan Bai, and et al. Autoregressive visual tracking. In *CVPR*, pages 9697–9706, 2023. 8

[82] Qiangqiang Wu, Tianyu Yang, and et al. Dropmae: Masked autoencoders with spatial-attention dropout for tracking tasks. In *CVPR*, pages 14561–14571, 2023. 1, 8

[83] Wanying Wu, Pengzhi Zhong, and Shuiwang Li. Fisher pruning for real-time uav tracking. In *IJCNN*, pages 1–7, 2022. 1, 2

[84] You Wu, Xucheng Wang, Dan Zeng, and et al. Learning motion blur robust vision transformers with dynamic early exit for real-time uav tracking. *arXiv preprint arXiv:2407.05383*, 2024. 1

[85] Fei Xie, Chunyu Wang, and et al. Learning tracking representations via dual-branch fully transformer networks. In *ICCV*, pages 2688–2697, 2021. 2

[86] Fei Xie, Chunyu Wang, Guangting Wang, and et al. Correlation-aware deep tracking. In *CVPR*, pages 8741–8750, 2022. 2

[87] Jinxia Xie and et al. Autoregressive queries for adaptive tracking with spatio-temporal transformers. In *CVPR*, pages 19300–19309, 2024. 6, 7

[88] Di Yang and et al. Self-supervised video pose representation learning for occlusion- robust action recognition. In *AFGR*, pages 1–5, 2021. 3

[89] Xiangyang Yang, Dan Zeng, and et al. Adaptively bypassing vision transformer blocks for efficient visual tracking. *Pattern Recognition*, 161:111278, 2025. 2

[90] Liangliang Yao, Changhong Fu, and et al. Sgdvit: Saliency-guided dynamic vision transformer for uav tracking. *arXiv preprint arXiv:2303.04378*, 2023. 6

[91] Botao Ye, Hong Chang, and et al. Joint feature learning and relation modeling for tracking: A one-stream framework. In *ECCV*, pages 341–357, 2022. 1, 2, 5, 7

[92] Sangdoo Yun, Dongyoon Han, and et al. Cutmix: Regularization strategy to train strong classifiers with localizable features. In *ICCV*, pages 6023–6032, 2019. 7

[93] Dan Zeng, Mingliang Zou, Xucheng Wang, and Shuiwang Li. Towards discriminative representations with contrastive instances for real-time uav tracking. In *ICME*, pages 1349–1354, 2023. 6

[94] Chenyuan Zhang, Jiu Xu, and et al. A klt-based approach for occlusion handling in human tracking. In *PCS*, pages 337–340, 2012. 3

[95] Yi Zhang, Pengliang Ji, and et al. 3d-aware neural body fitting for occlusion robust 3d human pose estimation. *ICCV*, pages 9365–9376, 2023. 3

[96] Haojie Zhao, Dong Wang, and Huchuan Lu. Representation learning for visual object tracking by masked appearance transfer. In *CVPR*, pages 18696–18705, 2023. 7

[97] Zikun Zhou, Wenjie Pei, Xin Li, and et al. Saliency-associated object tracking. In *ICCV*, pages 9846–9855, 2021. 7

[98] Pengfei Zhu, Longyin Wen, and et al. Visdrone-vdt2018: The vision meets drone video detection and tracking challenge results. In *ECCV Workshops*, 2018. 5, 6, 7, 8